Effects of common polymorphisms on the properties of recombinant human methylenetetrahydrofolate reductase

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Methylenetetrahydrofolate reductase (MTHFR) catalyzes the conversion of methylenetetrahydrofolate to methyltetrahydrofolate, the major methyl donor for the conversion of homocysteine to methionine. Two common polymorphisms of the human enzyme have been identified: 677C>T, which leads to the substitution of Ala-222 by valine, and 1298A>C, which leads to the replacement of Glu-429 by alanine; the former polymorphism is the most frequent genetic cause of mild hyperhomocysteinemia, a risk factor for cardiovascular disease. By using a baculovirus expression system, recombinant human MTHFR has been expressed at high levels and purified to homogeneity in quantities suitable for biochemical characterization. The Glu429Ala protein has biochemical properties that are indistinguishable from the wild-type enzyme. The Ala222Val MTHFR, however, has an enhanced propensity to dissociate into monomers and to lose its FAD cofactor on dilution; the resulting loss of activity is slowed in the presence of methyltetrahydrofolate or adenosylmethionine. This biochemical phenotype is in good agreement with predictions made on the basis of studies comparing wild-type *Escherichia coli* MTHFR with a mutant, Ala177Val, homologous to the Ala222Val mutant human enzyme [Guenther, B. D., *et al.* (1999) *Nat. Struct. Biol.* 6, 359–365].

Given the many polymorphisms being identified by genome sequencing or discovered in the course of genetic analysis, there is a critical need to establish whether a polymorphism affects physiological function before initiating clinical studies aimed at nutritional intervention or pharmacologic treatment. It remains extremely difficult to predict whether a mutation will alter function without biochemical analysis of the mutant protein. In the present paper, we compare the properties of recombinant human methylenetetrahydrofolate reductase (MTHFR) with those of mutant proteins corresponding to two common polymorphisms of MTHFR in humans.

MTHFR is a flavoprotein that catalyzes reduction of methylenetetrahydrofolate (CH_2-H_4folate) to methyltetrahydrofolate (CH_3-H_4folate) by using NAD(P)H as a reducing agent. CH_3-H_4folate serves as the methyl donor in the conversion of homocysteine to methionine catalyzed by cobalamin-dependent methionine synthase, and deficiencies in MTHFR activity result in elevated levels of homocysteine in the blood.

Mild elevation of blood homocysteine is an established risk factor for cardiovascular disease (1, 2) and is associated with increased mortality in patients with confirmed coronary artery disease (3). Mice homozygous for a knock-out mutation in MTHFR show 10-fold elevations in plasma homocysteine, and abnormal lipid deposition in the proximal portion of the aorta is observed in older heterozygotes and in homozygotes, but not in wild-type animals (4). Hyperhomocysteinemia is also linked to an increased incidence of neural tube defects in the human fetus (5). There is an inverse relationship between the level of folates in the blood and plasma homocysteine (1), presumably because folate deficiency leads to reduced levels of CH_3-H_4folate and decreased conversion of homocysteine to methionine by methionine synthase. For this reason, polymorphisms in human MTHFR have received considerable scrutiny as potential risk factors for the development of mild hyperhomocysteinemia and its attendant sequelae.

Patients with reduced MTHFR activity, due to a thermolabile form of the enzyme, were first identified by Kang *et al.* (6) and Rosenblatt and Erbe (7) and were found to be overrepresented in patients with cardiovascular disease (6). The isolation of a cDNA for human MTHFR permitted the identification of the 677C>T (Ala222Val) polymorphism (8), which was shown to be associated with increased thermolability of MTHFR activity in both heterozygotes and homozygotes (9). This polymorphism is a common genetic determinant of plasma homocysteine in the general population; ≈10–15% of Caucasians of European origin and of Japanese are homozygous for the polymorphism, whereas the incidence is higher in Mediterranean countries and lower in Afro-Americans (10). Homozygosity for the 677C>T polymorphism is associated with mild hyperhomocysteinemia in humans with plasma folate concentrations below the median (11) and is the most common genetic cause of mild elevations in plasma homocysteine in humans (10). Numerous studies have attempted to assess the correlation of the 677C>T polymorphism with cardiovascular disease (recently reviewed in ref. 12). No consistent correlation between homozygosity for the polymorphism and cardiovascular risk or longevity has been observed, but the studies published thus far do not have sufficient statistical power to detect the small risk enhancement related to a mean increment in total serum homocysteine of 2.6 μmol liter^{-1} detected in subjects with the TT as compared with the CC genotype (12). This variant has also been reported to alter risk for several other common disorders (10).

Mammalian MTHFRs are dimers of ≈70 kDa subunits; each subunit contains noncovalently bound FAD. The polypeptide contains an N-terminal catalytic domain and a C-terminal regulatory domain where the allosteric inhibitor AdoMet binds and regulates enzyme activity in response to the level of methionine in the cell (13). The N-terminal catalytic domain is homologous to a smaller MTHFR in *Escherichia coli*, which lacks the regulatory domain. Human MTHFR is expressed at very low levels in *E. coli* and in yeast; thus biochemical characterization of the phenotype associated with the 677C>T polymorphism in the purified enzyme has not been possible. However, the *E. coli* MTHFR has been overexpressed, and the structure of the wild-type enzyme has been determined (14). Ala-177, which is homologous to Ala-222 in human MTHFR, is located with its

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Abbreviations: CH_2-H_4folate, 5,10-methylenetetrahydrofolate; CH_3-H_4folate, 5-methyltetrahydrofolate; MTHFR, methylenetetrahydrofolate reductase.

See commentary on page 14754.

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side chain on the inside of a tight loop between helix $\alpha 5$ and strand $\beta 6$ at the bottom of an $\alpha_8\beta_8$ barrel. This positions the site of mutation far from the active site. However, the tight loop could not accommodate the side chain of a valine without distortion, and it was hypothesized that the polymorphism would lead to movement of helix $\alpha 5$, which has numerous interactions with the FAD cofactor. The properties of the Ala177Val mutant MTHFR were compared with those of the wild-type enzyme; the mutation did not affect the kinetics of the reduction of CH_2-H_4folate to CH_3-H_4folate but it did lead to decreased affinity of the enzyme for its FAD cofactor (14).

There are several important differences between the mammalian MTHFRs and those from *E. coli*. The prokaryotic enzyme is a tetramer of 33-kDa subunits, whereas the mammalian enzymes are dimers of ≈70-kDa subunits. Because FAD loss from the *E. coli* enzyme is associated with conversion of the tetrameric enzyme into dimers, the different oligomeric state of the mammalian enzymes might affect the propensity for flavin dissociation. Furthermore, the prokaryotic enzyme uses NADH as a reductant whereas the mammalian enzymes use NADPH, and the prokaryotic enzyme lacks the regulatory domain entirely.

The 1298A>C polymorphism leads to the substitution of Glu-429 in the regulatory domain of the human enzyme by an alanine (15, 16). Although the original cDNA of human MTHFR contained a C at nucleotide 1298 (9), A is the more common allele. The incidence of the CC genotype at nucleotide 1298 is similar to that of the TT genotype at nucleotide 677 in those populations that have been studied, ≈10% in Caucasians of Northern European origin. The 1298A>C mutation is associated with a mild reduction in MTHFR activity *in vivo* (15, 16) and *in vitro* (17), and has been suggested to be associated with increased risk for neural tube defects (15).

In the present study, we compare the properties of wild type, Ala222Val, Glu429Ala, and Ala222Val Glu429Ala homogeneous recombinant human MTHFR proteins. Our studies fail to distinguish between the properties of the Glu429Ala mutant protein and the wild-type protein, or between the properties of the Ala222Val mutant protein and the Ala222Val Glu429Ala double mutant. However, the properties of the Ala222Val mutant are clearly distinguishable from the wild-type enzyme and remarkably resemble the properties predicted in model studies of the effect of the homologous Ala177Val mutation in *E. coli* MTHFR.

Methods

Generation of Baculovirus Containing Recombinant Human MTHFR. The N-terminal sequence of MTHFR purified from human tissue has not yet been determined; therefore, the actual start site of translation of the human enzyme is not known. Several potential start sites have been identified. On the basis of similarity to Kozak's sequence (18), we chose nucleotide 13 in the human MTHFR sequence (GenBank accession no. U09806) as the probable start site. A 2.2-kb fragment containing the coding region of the human MTHFR cDNA was obtained from plasmid pTrc99AMR (8) by digestion with *Nco*I and *Xba*I. The gene in plasmid pTrc99AMR contains the C allele at nucleotide 1298, and PCR-based mutagenesis was used to construct a fragment containing the A allele (17). The resulting fragment was introduced into donor vectors pFastBac1 and pFastBacHT (GIBCO/BRL) to express wild-type untagged and N-terminally histidine-tagged proteins, respectively. C1298, T677, and C1298 T677 variants were introduced into plasmid pFastBacHT. Each MTHFR cDNA sequence was verified by sequencing and by PCR-based restriction fragment length polymorphism analysis (9, 10) with modified primer pairs, after the construction of each donor vector. Recombinant baculovirus were generated by using the Bac to Bac expression system (GIBCO/BRL) according to the manufacturer's protocol.

Maintenance of Insect Cells and Expression of Human MTHFR. Sf9 cells from *Spodoptera frugiperda* were cultured at 27°C in Sf-900II SFM medium (GIBCO/BRL) containing gentamicin (10 μg/liter) and amphotericine B (250 μg/liter). Suspension cultures (0.6 liter) were grown in an incubator shaker at 130 rpm. Amplified baculovirus stock was added at a cell density of $\approx 1.5 \times 10^6$ cells per ml. The infected cells were harvested ≈72 h postinfection. After washing with PBS, cells were collected and stored at −80°C.

MTHFR Assay. Activities were determined by using three different assays. NADPH-menadione oxidoreductase activity was determined as described (19), except that the assay volume was reduced to 1 ml. CH_3-H_4folate-menadione oxidoreductase activity was determined as described (20), except that the assay volume was reduced to 250 μl. The physiological NADPH-CH_2-H_4folate oxidoreductase activity was determined by modification of a described protocol (21). Potassium phosphate buffer (50 mM, pH 6.7/100 μM NADPH) and enzyme in a total volume of 1 ml were mixed in a cuvette equilibrated at 25°C, and the reaction was started by addition of CH_2-H_4folate. Absorbance changes were monitored at 340 nm. (6*R*)CH_2-H_4folate was generated by addition of a 5-fold excess of formaldehyde to a solution of (6*S*)H_4folate (Eprova, Schaffhausen, Switzerland) in 0.25 M triethanolamine chloride buffer (pH 7.0) (500 mM 2-mercaptoethanol) that had been equilibrated with argon.

Purification of Histidine-Tagged MTHFR. Pooled infected insect cells, 10 g from a 1.5-liter culture, were homogenized by sonication in 40 ml of 50 mM potassium phosphate buffer (pH 7.2) containing 0.1 M NaCl and phenylmethylsulfonylfluoride (1 mM). The homogenate was centrifuged at 30,000 × *g* for 30 min at 4°C, and the supernatant was added to a 20-g cake of DEAE cellulose that had been equilibrated with 50 mM potassium phosphate buffer (pH 7.2) containing 0.1 M NaCl and then filtered in a Buchner funnel to remove excess buffer. The slurry was incubated at 4°C for 30 min and then packed into a 2.5-cm diameter column. The column was washed with 3 volumes of the equilibrating buffer, and then the enzyme was eluted with 2 volumes of 50 mM potassium phosphate buffer (pH 7.2) containing 0.3 M NaCl. The enzyme eluted as a distinct yellow band, and activity was monitored by using the NADPH-menadione oxidoreductase assay. The active yellow fractions were pooled, and 2 M imidazole (pH 7.5) was added to give a final concentration of 0.1 M imidazole. The pooled fractions were applied to a 1-ml Hi-Trap nickel-affinity column (Amersham Pharmacia) that had been equilibrated with 50 mM potassium phosphate buffer (pH 7.2) containing 0.3 M NaCl and 0.1 M imidazole. The column was washed with the same buffer, and the His-tagged MTHFR was eluted by using 50 mM potassium buffer (pH 7.2) containing 0.3 M NaCl, 0.3 M imidazole, and 10% glycerol. NaCl and imidazole were removed by gel filtration, and the protein was stored at −80°C.

Measurement of FAD Release. A FluoroMax-2 spectrofluorometer (JY/Spex Industries, Edison, NJ) was used to determine the fluorescence of FAD. The excitation wavelength was 390 nm, and emission was monitored at 525 nm. A concentrated stock of enzyme (>100 μM) was diluted into 50 mM phosphate buffer (pH 7.2) containing 0.3 mM EDTA and 10% glycerol to initiate flavin release. The enzyme-bound flavin is completely quenched, and the fluorescence increases as FAD is released. Although typically only a single data point is shown for each experiment, experiments were repeated at least twice, and comparable results were obtained at each repetition.

Table 1. Purification of histidine-tagged human MTHFR

Step	Total activity,* μmol min^{-1}	Specific activity,* μmol min^{-1} mg^{-1}†	Purification-fold	Yield, %
Crude extract	96.2	0.20	1.0	100
DEAE cellulose	67.1	0.93	4.6	69.7
Nickel affinity	43.5	12.4	62	45.2

Approximately 10 g of wet weight of cells, harvested from a 1.5-liter culture, were used.

*Enzyme activity was determined by using the CH_3-H_4folate:menadione oxidoreductase assay.

†The concentration of protein was determined by using the BioRad protein assay kit, or for purified enzyme, by measuring the FAD absorbance using a molar extinction coefficient of 12,100 M^{-1}cm^{-1} at 450 nm (22).

Measurement of Thermolability. The test for thermolability of enzyme in crude extracts has been described (14). After heat treatment, the enzyme was assayed with the CH_3-H_4folate-menadione oxidoreductase assay. Purified enzyme was incubated for 15 min at 46°C in the presence of varying amounts of FAD and/or (6*R,S*)CH_3-H_4folate (barium salt, Sigma-Aldrich) before assay.

Results

Expression and Purification of Human MTHFR. Histidine-tagged MTHFR was readily purified to homogeneity by using a nickel-affinity column (Table 1). The specific activity of 12.4 μmol min^{-1} mg^{-1} is slightly lower than the value of 19.4 μmol min^{-1} mg^{-1} obtained for the homogenous porcine enzyme (19), but is similar to the specific activity of homogeneous untagged recombinant human enzyme (data not shown). Approximately 3.5 mg of purified enzyme was obtained from 10 g of wet cells harvested from a 1.5-liter suspension culture. Despite the fact that FAD was not added to the buffers during purification, the purified enzyme was replete with FAD as judged by the ratio of FAD/cysteine thiols in the protein. However, the purification protocol was designed to keep the enzyme at high concentration throughout the procedure. The histidine-tagged mutant enzymes could be purified to homogeneity by using the same procedure, and yielded enzymes of comparable specific activity.

Steady-State Kinetics. Steady-state kinetic analyses of wild-type and Ala222Val, Glu429Ala, and Ala222Val Glu429Ala mutant histidine-tagged recombinant human MTHFRs using the NADPH-CH_2-H_4folate oxidoreductase assay revealed no significant differences in the K_m values for NADPH or CH_2-H_4folate or in V_{max} (Fig. 6, which is published as supporting information on the PNAS web site, www.pnas.org). We also failed to see differences between wild-type and mutant enzymes in the kinetic parameters associated with the NADH-menadione and CH_3-H_4folate-menadione oxidoreductase assays. The activities of all four enzymes were strongly inhibited by AdoMet (Fig. 7, which is published as supporting information on the PNAS web site), and significant differences in the K_i values for AdoMet were not observed. Thus, these mutations do not affect catalytic function or AdoMet regulation *in vitro*.

FAD Release. Studies with MTHFR from *E. coli* had established that the Ala177Val mutation, which is homologous to Ala222Val in human MTHFR, yields an enzyme that loses its FAD cofactor on dilution about 11 times more rapidly than the wild-type enzyme (14). Because the flavin is essential for transferring electrons between NADH and CH_2-H_4folate, flavin release is accompanied by loss of activity. The initial rate of flavin loss varies as the square root of the subunit concentration of the enzyme after dilution, indicating that dissociation of the tetramer into dimers precedes flavin loss. Fig. 1*A* shows the results of similar experiments performed with human wild-type and Ala222Val MTHFR. Ala222Val enzyme releases FAD ≈3-fold faster than wild-type enzyme. In Fig. 1*B*, the initial rate of flavin loss is plotted against the subunit concentration of the enzyme after dilution for each of the four MTHFR variants. Each plot shows that the rate of flavin loss varies as the square root of the subunit concentration after dilution, indicating that in each case, dissociation of the dimeric enzyme into monomers precedes flavin loss. Scheme 1, which is published as supporting infor-

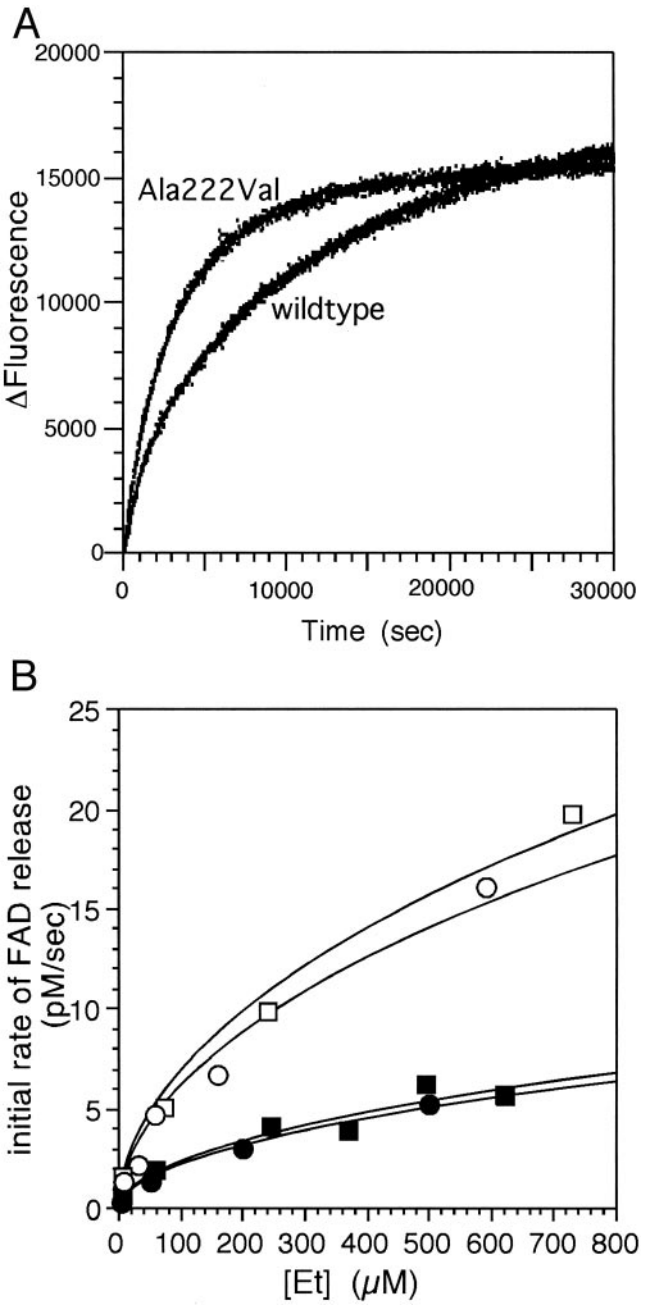

Fig. 1. FAD dissociation after dilution of wild-type and mutant enzymes. Released FAD was detected by fluorometry. The enzyme solution was incubated at 37°C. (*A*) The change in fluorescence over time is shown for wild-type and Ala222Val mutant enzyme diluted to 50 nM. (*B*) The initial rate of release of FAD is plotted against enzyme concentration after dilution for the wild-type and mutant enzymes: wild type (●), Ala222Val (○), Glu429Ala (■), and Ala222Val Glu429Ala (□).

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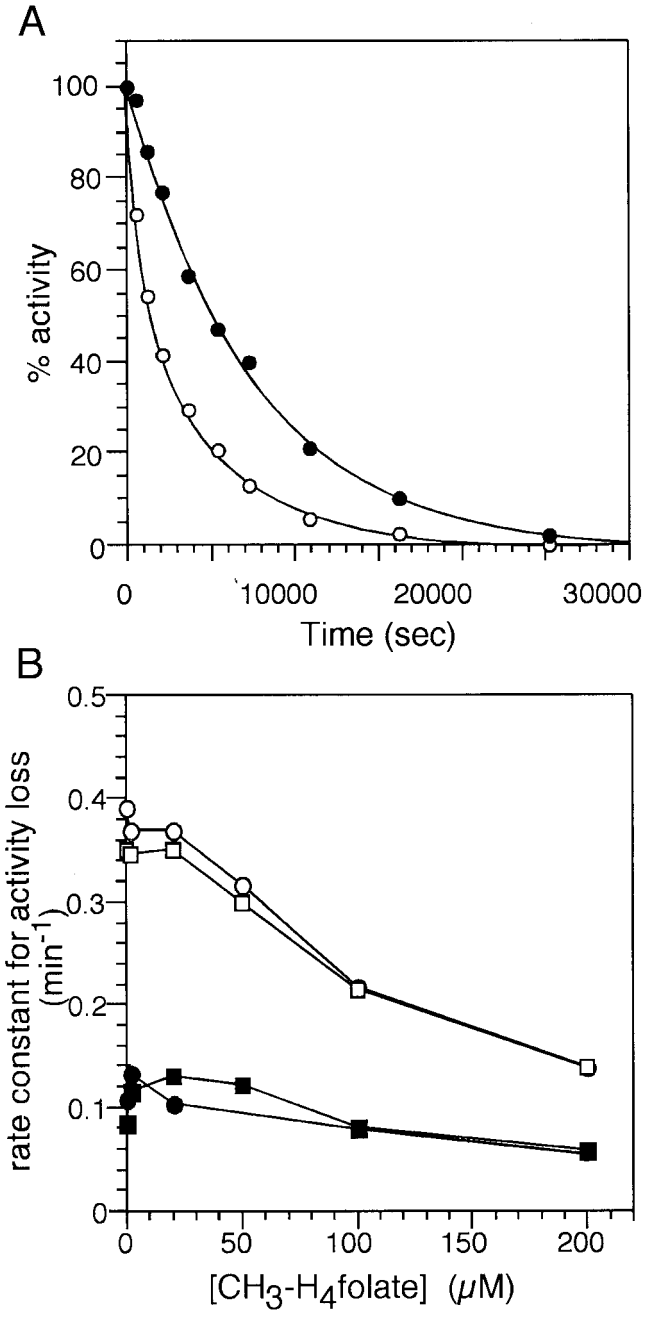

Fig. 2. Protection of activity loss after dilution in the presence of CH_3-H_4folate. Enzyme was incubated with (6*S*)CH_3-H_4folate (Eprova) at 37°C. The residual enzyme activity after dilution was determined by CH_3-H_4folate:menadione oxidoreductase assay. (*A*) The time course of activity loss of Ala222Val mutant enzyme with (●) or without (○) 200 μM CH_3-H_4folate. (*B*) The rate constant for initial activity loss of wild-type (●), Ala222Val (○), Glu429Ala (■), and Ala222Val Glu429Ala (□) mutant enzymes is plotted against the concentration of CH_3-H_4folate present after dilution. The enzyme concentration after dilution is 50 nM.

mation on the PNAS web site, contains a derivation for this relationship. The rate of flavin dissociation from the Glu429Ala mutant is indistinguishable from that of the wild-type enzyme, whereas the rate of flavin dissociation from the Glu429Ala Ala222Val double mutant cannot be distinguished from that for the Ala222Val mutant.

For these studies, the enzyme was diluted to 50 nM subunit concentration. A rough calculation of the intracellular enzyme

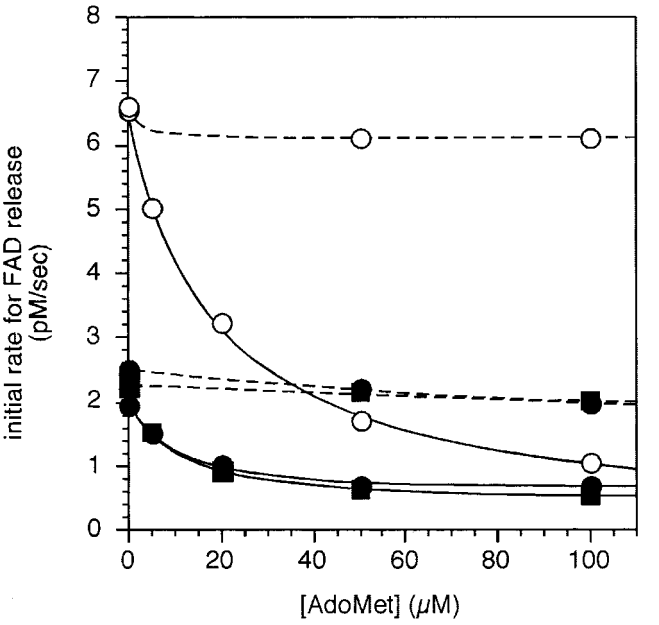

Fig. 3. Effect of AdoMet and AdoHcy on the initial rate of FAD release after dilution. Conditions are similar to those in Fig. 2. The initial rate of release of FAD is plotted against AdoMet concentration after dilution for the wild-type and mutant enzymes: wild type (●), Glu429Ala (■), and Ala222Val (○). The solid lines are fitted to data obtained in the absence of AdoHcy, and the dashed lines are fitted to data obtained in the presence of 100 μM AdoHcy and the indicated concentrations of AdoMet. Data for Ala22Val and for the double Ala222Val Glu429Ala mutant were indistinguishable (data not shown).

concentration in porcine liver can be derived from the purification data for porcine MTHFR (22) and yields a value of ≈36 nM enzyme. Therefore, we believe that the *in vitro* experiments shown in Fig. 1 are pertinent to the *in vivo* conditions, and that substantial concentrations of inactive monomeric apoenzyme may contribute to the decreased activity, ≈30% of that in humans with the CC genotype (9), seen in the cell extracts from humans homozygous for the polymorphism.

CH_3-H_4Folate Protects Human MTHFR from Loss of FAD After Dilution. (6*S*)CH_3-H_4folate (Eprova) slows the rate of FAD dissociation in a dose-dependent fashion after dilution of *E. coli* MTHFR (14). The effect is seen with both wild-type enzyme and the Ala177Val mutation, although protection is more dramatic with the latter enzyme. The protective effect of 200 μM CH_3-H_4folate on activity loss after dilution of human wild-type and Ala222Val MTHFR is shown in Fig. 2*A*. In Fig. 2*B*, plots of the initial rate constant for activity loss against the concentration of CH_3-H_4folate are shown for all four enzyme variants. Again, the results for the Glu429Ala mutant are indistinguishable from those of the wild-type enzyme, both in the presence and in the absence of the Ala222Val mutation. This figure demonstrates the graded effect of increasing CH_3-H_4folate concentration on the initial rate of activity loss after dilution. For these *in vitro* studies, we have used the commercially available monoglutamate form of CH_3-H_4folate, which binds with relatively low affinity to the enzyme. The intracellular forms of CH_3-H_4folate are almost exclusively polyglutamates, which bind with substantially higher affinity and which would be expected to protect the enzyme under more physiologically relevant conditions. Addition of 50 μM FAD, a concentration approximately twice that of intracellular FAD in rat liver (23), also protects against loss of activity after dilution (data not shown).

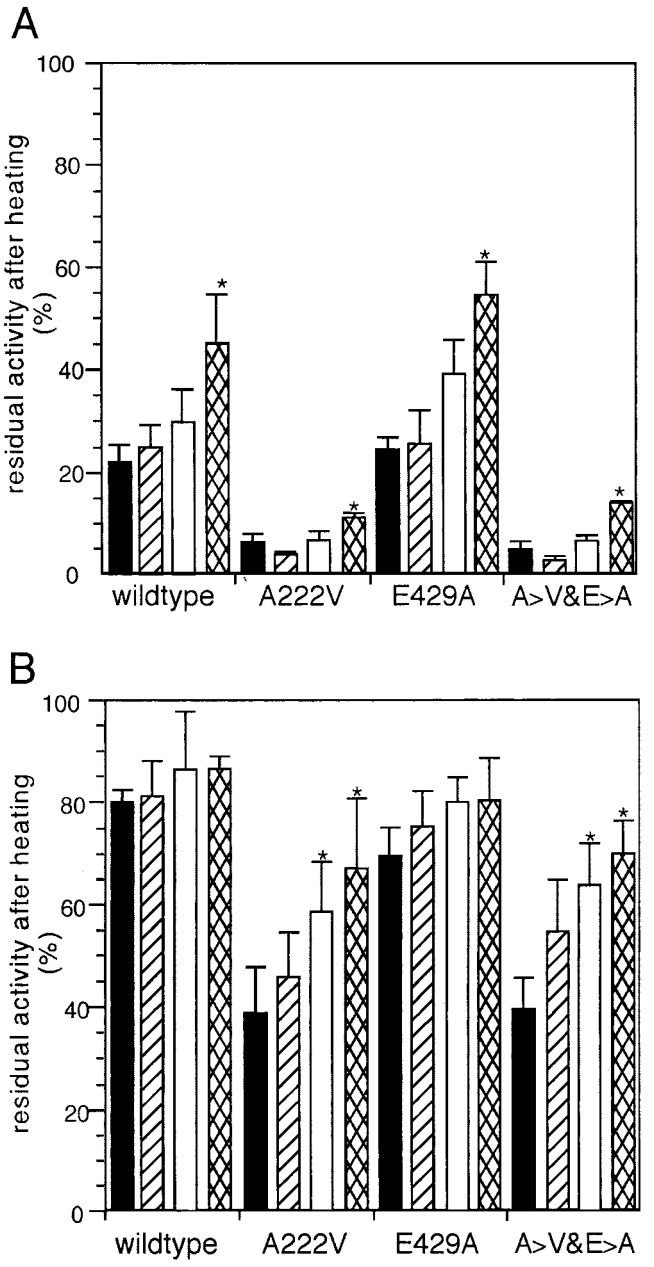

Fig. 4. Protection against heat inactivation by (6-*R, S*)CH_3-H_4folate with or without added FAD. Purified wild-type, Ala222Val (A222V), Glu429Ala (E429A), and Ala222Val Glu429Ala (A>V & E>A) mutant MTHFRs were preincubated at 46°C in the presence of 0, 120, 240, or 360 μM (6-*RS*)CH_3-H_4folate without (*A*) or with (*B*) 10 μM FAD. After cooling to 37°C, the residual enzyme activity was determined by CH_3-H_4folate:menadione oxidoreductase assay. The values are expressed as mean ± SD of data obtained from three experiments. Statistically significantly differences ($P < 0.03$) from the activity of the same variant in the absence of CH_3-H_4folate are indicated by an asterisk. The different concentrations of CH_3-H_4folate are represented by the fill pattern of the bars: 0, black; 120, hatched; 240, white; and 360 μM, checkered.

AdoMet Protects Human MTHFR from Loss of FAD After Dilution and AdoHcy Blocks the Effect of AdoMet. Fig. 3 shows the protective effect of AdoMet (chloride salt, Sigma) in decreasing the rate of FAD release after MTHFR dilution and reveals that AdoMet has two divergent roles in regulating enzyme activity. It serves as a reversible allosteric inhibitor of enzyme activity (13) while stabilizing the enzyme by slowing FAD dissociation. AdoHcy (Sigma) competes with AdoMet for binding to the regulatory domain, but neither activates nor inhibits activity, thus it prevents inhibition by AdoMet. As shown here, AdoHcy also reverses the protective effect of AdoMet. Thus, we expect the stability of the enzyme to be affected by the AdoMet/AdoHcy ratio in a manner inverse to the activity of the enzyme.

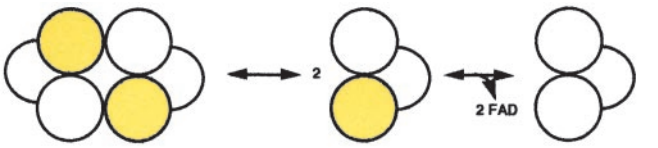

Fig. 5. Proposed mechanism for FAD dissociation. Each subunit consists of a catalytic (yellow) and a regulatory (white) domain, connected by a linker. The homodimer dissociates into monomers in a rapid and reversible step that is followed by rate-limiting FAD release.

Thermolability. Purified mutant and wild-type enzymes were examined by using the simple assay for thermolability that was widely used to detect the Ala222Val mutation in lymphocyte extracts before the development of restriction fragment length polymorphism analysis. Fig. 4 shows the effects of added CH_3-H_4folate on residual activity of the purified enzyme variants after heating to 46°C for 15 min in the absence (Fig. 4*A*) or presence (Fig. 4*B*) of added FAD. The thermolability of the Ala222Val enzyme as compared with the wild-type and Glu429Ala proteins is obvious in Fig. 4 *A* and *B*, and it is clear that the Glu429Ala mutation does not affect the thermolability of either the wild-type or Ala222Val proteins. These results can reasonably be interpreted as suggesting that thermolability is enhanced when the FAD cofactor dissociates, and compounds that slow the rate of flavin loss also protect the enzyme against thermal inactivation. The thermolability assay thus serves as a surrogate for the effect of flavin loss on enzyme activity in the cellular milieu, where MTHFR is present in very low concentration.

Discussion

Use of a baculovirus expression system has, for the first time to our knowledge, allowed expression of recombinant human MTHFR at levels suitable for biochemical characterization. The human enzyme has properties that are generally similar to the well characterized porcine MTHFR (24). However, comparison of the phenotypes conferred by polymorphisms in the human enzyme requires a genetically manipulable expression system, such as the one used here. These comparisons fail to distinguish the phenotype of the purified 1298A>C mutant protein (Glu429Ala) from that of its wild-type allele, whether in the presence or the absence of the 677C>T mutation. Our results contrast with the decreased activity levels seen for this mutant in lymphocyte extracts. Decreased levels of activity have also been seen in crude extracts when the recombinant human enzyme is expressed in bacteria (17). Taking all these results into account, we suggest that the decreased levels of activity stem from differences in protein stability rather than from the properties of the purified enzyme *per se*.

Our characterization of the properties of the 677C>T (Ala222Val) mutation are generally consistent with earlier studies with the *E. coli* enzyme (14), suggesting the validity of such types of model studies. However, the phenotypic differences between wild-type and mutant human enzymes are more subtle, consistent with retention of the allele in the human population.

Particularly interesting is the dissociation of the enzyme after dilution and preceding flavin release. The prokaryotic enzyme is a tetramer of identical catalytic domains and dissociates into dimers before release of FAD (14). The x-ray structure reveals an unusual structure for the tetramer, with four subunits ar-

ranged in a planar rosette and only 2-fold rotational symmetry. Each subunit interacts extensively with one of its neighbors and much less with the other, favoring dissociation of the tetramer into dimers rather than monomers (14). The mammalian enzymes are dimers, and scanning transmission electron microscopy reveals that they too form planar rosettes with four discrete domains (25). By using the bacterial enzyme as a model, we would predict a head to tail arrangement of the monomers, with each catalytic domain flanked by regulatory domains in the planar rosette (Fig. 5). In this context, our observation that dissociation of the enzyme into monomers precedes flavin release suggests that allosteric interactions between the regulatory domains of one subunit and the catalytic domain of another modulate the affinity for the flavin cofactor. AdoMet, which binds to the regulatory domain (13), clearly influences the affinity of the catalytic domain for its flavin cofactor, suppressing the release of flavin after dilution.

The effect of AdoMet on cofactor binding, and its reversal by AdoHcy, is potentially significant for human physiology. The intracellular concentrations of AdoMet ($\approx$100 μM) and AdoHcy ($\approx$50 μM) measured in rat liver (26) are comparable to those we used *in vitro*. TT Homozygotes for the 677C>T polymorphism have elevated plasma homocysteine when their folate status is low, and in the heterozygote MTHFR knockout mouse, which has similar MTHFR activity, the elevated homocysteine is associated with decreased AdoMet/AdoHcy ratios in all tissues (4). Thus, when the folate status of a TT homozygote is low, the decreased flux of homocysteine into methionine associated with destabilization of MTHFR will compound the stability problem and lead to further FAD release and loss of MTHFR activity. In contrast to the reversible enzyme inhibition induced by AdoMet, flavin dissociation and the associated loss of enzyme activity when the AdoMet/AdoHcy ratio is low is likely to be irreversible *in vivo*, due to instability of the apoenzyme. Our studies suggest that TT homozygotes will be particularly at risk when their folate status is low.

After the elucidation of the phenotype of the Ala177Val mutation in *E. coli* MTHFR, which revealed that flavin dissociation might be a key feature of the 677C>T phenotype in humans, a number of laboratories have demonstrated a correlation between hyperhomocysteinemia and riboflavin deficiency (27). In the rat, MTHFR activity is particularly sensitive to riboflavin deficiency (23). Our studies with the human enzyme suggest that folate, methionine, and riboflavin status will all be important for the maintenance of MTHFR activity, and that TT homozygotes will also be at risk when their riboflavin status is low.

Given the deleterious sequelae associated with elevated homocysteine in TT homozygotes, and the biochemical phenotype associated with this mutation, one may question why the allele persists at such high levels in human populations. The most plausible rationale comes from studies showing a correlation between low folate status and uracil misincorporation into DNA and chromosome breakage (28). These studies suggested that, when folate status is low, the inactivation of MTHFR by flavin dissociation might spare folate in the methylenetetrahydrofolate form needed for conversion of dUMP into dTMP by disabling the conversion of methylene- to methyltetrahydrofolate. The 677C>T mutation modulates the set point of inactivation. Consistent with this argument are several reports suggesting that the common variant is protective in certain neoplasias, including colon cancer and leukemia (12, 29). The enhanced conversion of dUMP into dTMP when folate levels are low could reduce chromosomal breakage or increase dTMP levels to allow DNA repair. Studies to assess the affect of the 677C>T polymorphism on uracil misincorporation and chromosome breakage will be essential to examine this hypothesis.

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1. Boushey, C. J., Beresford, S. A. A., Omenn, G. S. & Motulsky, A. G. (1995) *J. Am. Med. Assn.* **274,** 1049–1057.
2. Graham, I. M., Daly, L. E., Refsum, H. M., Robinson, K., Brattstrom, L. E., Ueland, P. M., Palma-Reis, J., Boers, G. H. J., Sheahan, R. G., Israelsson, B., *et al.* (1997) *J. Am. Med. Assn.* **277,** 1775–1781.
3. Refsum, H., Ueland, P. M., Nygard, O. & Vollset, S. E. (1998) *Annu. Rev. Med.* **49,** 31–62.
4. Chen, Z., Karaplis, A. C., Ackerman, S. L., Pobribny, I. P., Melnyk, S., Lussier-Cacan, S., Chen, M. F., Pai, A., John, S. W. M., Smith, R. S., *et al.* (2001) *Hum. Mol. Genet.* **10,** 433–443.
5. Shields, D. C., Kirke, P. N., Mills, J. L., Ramsbottom, D., Molloy, A. M., Burke, H., Weir, D. G., Scott, J. M. & Whitehead, A. S. (1999) *Am. J. Hum. Genet.* **64,** 1045–1055.
6. Kang, S. S., Passen, E. L., Ruggie, N., Wong, P. W. & Sora, H. (1993) *Circulation* **88,** 1463–1469.
7. Rosenblatt, D. S. & Erbe, R. W. (1977) *Pediatr. Res.* **11,** 1141–1143.
8. Goyette, P., Sumner, J. S., Milos, R., Duncan, A. M. V., Rosenblatt, D. S., Matthews, R. G. & Rozen, R. (1994) *Nat. Genet.* **7,** 195–200.
9. Frosst, P., Blom, H. J., Milos, R., Goyette, P., Sheppard, C. A., Matthews, R. G., Boers, G. J., den Heijer, M., Kluijtmans, L. A., van den Heuvel, L. P., *et al.* (1995) *Nat. Genet.* **10,** 111–113.
10. Schwahn, B. & Rozen, R. (2001) *Am. J. Pharmacogenom.* **1,** 189–201.
11. Jacques, P. F., Bostom, A. G., Williams, R. R., Ellison, R. C., Eckfeldt, J. H., Rosenberg, I. H., Selhub, J. & Rozen, R. (1996) *Circulation* **93,** 7–9.
12. Ueland, P. M., Hustad, S., Schneede, J., Refsum, H. & Vollset, S. E. (2001) *Trends Pharm. Sci.* **22,** 195–201.
13. Sumner, J., Jencks, D. A., Khani, S. & Matthews, R. G. (1986) *J. Biol. Chem.* **261,** 7697–7700.
14. Guenther, B. D., Sheppard, C. A., Tran, P., Rozen, R., Matthews, R. G. & Ludwig, M. L. (1999) *Nat. Struct. Biol.* **6,** 359–365.
15. van der Put, N. M., Gabreels, F., Stevens, E. M., Smeitink, J. A., Trijbels, F. J., Eskes, T. K., van den Heuvel, L. P. & Blom, H. J. (1998) *Am. J. Hum. Genet.* **62,** 1044–1051.
16. Weisberg, I., Tran, P., Christensen, B., Sibani, S. & Rozen, R. (1998) *Mol. Genet. Metab.* **64,** 169–172.
17. Weisberg, I. S., Tran, P., Christensen, B., Sibani, S. & Rozen, R. (2001) *Atherosclerosis* **156,** 409–415.
18. Kozak, M. (1984) *Nucleic Acids Res.* **12,** 857–872.
19. Matthews, R. G. (1986) *Methods Enzymol.* **122,** 372–381.
20. Sheppard, C. A., Trimmer, E. E. & Matthews, R. G. (1999) *J. Bacteriol.* **181,** 718–725.
21. Matthews, R. G. & Haywood, B. J. (1979) *Biochemistry* **18,** 4845–4851.
22. Daubner, S. C. & Matthews, R. G. (1982) *J. Biol. Chem.* **257,** 140–145.
23. Bates, C. J. & Fuller, N. J. (1986) *Brit. J. Nutr.* **55,** 455–464.
24. Matthews, R. G. (1991) in *Chemistry and Biochemistry of Flavoenzymes*, ed. Müller, F. (CRC, Boca Raton, FL), Vol. I, pp. 371–387.
25. Matthews, R. G., Vanoni, M. A., Khani, S., Hainfeld, J. F. & Wall, J. (1984) in *Flavins and Flavoproteins*, eds. Bray, R. C., Engel, P. C. & Mayhew, S. G. (Walter de Gruyter, Berlin), pp. 217–220.
26. She, Q.-B., Nagao, I., Hayakawa, T. & Tsuge, H. (1994) *Biochem. Biophys. Res. Commun.* **205,** 1748–1754.
27. Hustad, S., Ueland, P. M., Vollset, S. E., Zhang, Y., Bjørke-Monsen, A. L. & Schneede, J. (2000) *Clin. Chem.* **46,** 1065–1071.
28. Blount, B. C., Mack, M. M., Wehr, C. M., MacGregor, J. T., Hiatt, R. A., Wang, G., Wickramasinghe, S. N., Everson, R. B. & Ames, B. N. (1997) *Proc. Natl. Acad. Sci. USA* **94,** 3290–3295.
29. Chen, J., Giovannuci, E. L. & Hunter, D. J. (1999) *J. Nutr.* **129,** 560S–564S.